

HELM COMMANDS CHEATSHEET

(A Complete Quick-Reference Guide)



1. Notation & Conventions

- `<angle-brackets>` : placeholders (e.g., `<release>`, `<chart>`)
- `[square-brackets]` : optional flags
- `-n <namespace>` : specify Kubernetes namespace
- `--values / -f` : custom values file
- `--set key=value` : set values from CLI
- `--dry-run` : simulate install/upgrade
- `--debug` : detailed output
- `--version` : chart version

2. Basics & Setup

- `helm version` – Show Helm version
- `helm env` – Show Helm environment info
- `helm help` – Helm help
- `helm completion bash` – Enable bash autocomplete

3. Repository Management

Manage Helm chart repositories

- `helm repo list` – List repositories
- `helm repo add <name> <url>` – Add repository
- `helm repo remove <name>` – Remove repository
- `helm repo update` – Update repo index
- `helm search repo <keyword>` – Search charts in repos

4. Chart Management

Work with Helm charts

- `helm create <chart-name>` – Create new chart structure
- `helm lint <chart>` – Lint a chart
- `helm package <chart>` – Package chart
- `helm pull <repo/chart>` – Download chart
- `helm pull <repo/chart> --untar` – Download & extract chart
- `helm show chart <chart>` – Show chart metadata
- `helm show values <chart>` – Show default values

5. Install & Deploy Charts

Deploy applications to Kubernetes

- `helm install <release> <chart>` – Install chart
- `helm install <release> <chart> -n <ns>` – Install in namespace
- `helm install <release> <chart> -f values.yaml` – Custom values
- `helm install <release> <chart> --set key=value` – Set values
- `helm install <release> <chart> --dry-run --debug` – Test install

6. Release Management

Manage installed releases

- `helm list` – List releases (default namespace)
- `helm list -A` – List releases in all namespaces
- `helm status <release>` – Release status
- `helm get all <release>` – Get all release info
- `helm get values <release>` – Get user-supplied values

7. Upgrade & Rollback

Update running applications safely

- `helm upgrade <release> <chart>` – Upgrade release
- `helm upgrade <release> <chart> -f values.yaml` – Upgrade with values
- `helm upgrade <release> <chart> --install` – Install if not exists
- `helm history <release>` – Release revision history
- `helm rollback <release> <revision>` – Rollback to revision

8. Uninstall & Cleanup

Remove Helm releases

- `helm uninstall <release>` – Uninstall release
- `helm uninstall <release> -n <namespace>` – Uninstall from namespace
- `helm list --uninstalled` – Show uninstalled releases

9. Values & Templates

Work with Helm templating

- `helm template <release> <chart>` – Render templates locally
- `helm template <chart> -f values.yaml` – Render with values
- `helm get manifest <release>` – Get rendered manifests

10. Dependencies

Manage chart dependencies

- `helm dependency list <chart>` – List dependencies
- `helm dependency update <chart>` – Update dependencies
- `helm dependency build <chart>` – Build dependencies

11. Plugins

Extend Helm functionality

- `helm plugin list` – List plugins
- `helm plugin install <url>` – Install plugin
- `helm plugin remove <name>` – Remove plugin

12. Debugging & Troubleshooting

Diagnose Helm issues

- `helm install --debug --dry-run` – Debug installation
- `helm upgrade --debug --dry-run` – Debug upgrade
- `helm lint <chart>` – Validate chart
- `kubectl describe pod <pod>` – Pod-level debugging
- `kubectl logs <pod>` – View logs

13. Security & Best Practices

- Use values.yaml instead of hardcoding
- Avoid secrets in plain text
- Use helm lint before deploy
- Version control Helm charts

14. Helm Workflow (One-Line Summary)

`repo → search → create → lint → install → upgrade → rollback → uninstall`